

Applied A.I. Solutions

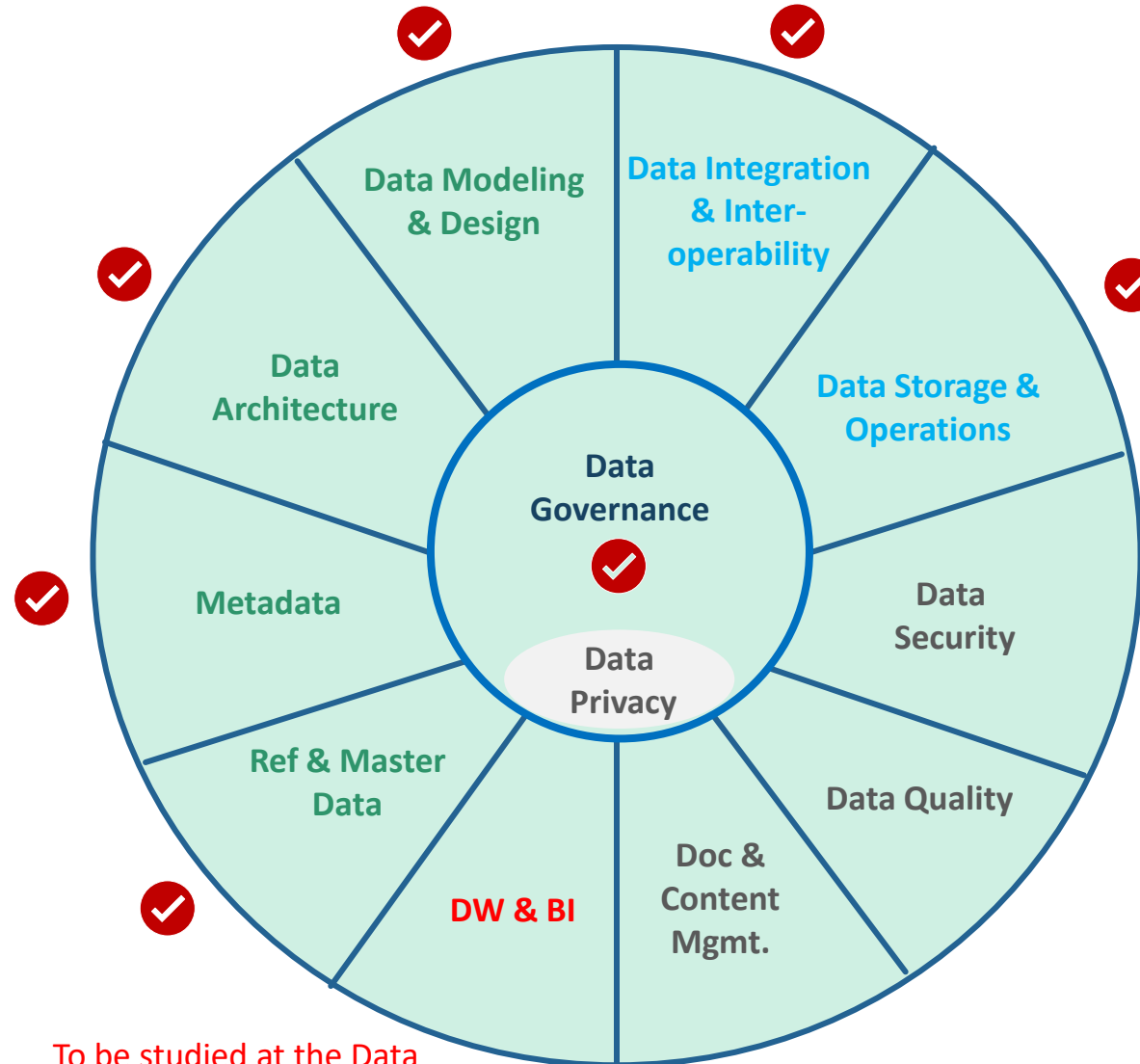
Foundations of Data Management

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DATA STORAGE & OPERATIONS

The DAMA Wheel



To be studied at the Data Visualization Techniques course

INTRODUCTION

- DS&O includes the **design, implementation and support of stored data**, to maximize its value through its lifecycle, from creation/acquisition to disposal
 - Database support to data life cycle
 - Data technology support

Data Storage and Operations Management Framework

Definition

The design, implementation, and support of stored data to maximize its value

DS&O Programs are guided by the following principles:

1. Plan for business **continuity**
2. Identify and act on **automation** opportunities
3. Build with **reuse** in mind
4. Understand and appropriately apply **best practices**
5. Utilize database **standards** to support requirements
6. Set expectations for the **DBA** role in project work

Goals

1. Ensure Business Continuity - minimize the risk of disruption
2. Manage **availability** of data throughout the data lifecycle
3. Ensure the **integrity** of data assets
4. Manage **performance** of data transactions



Business Drivers

Inputs

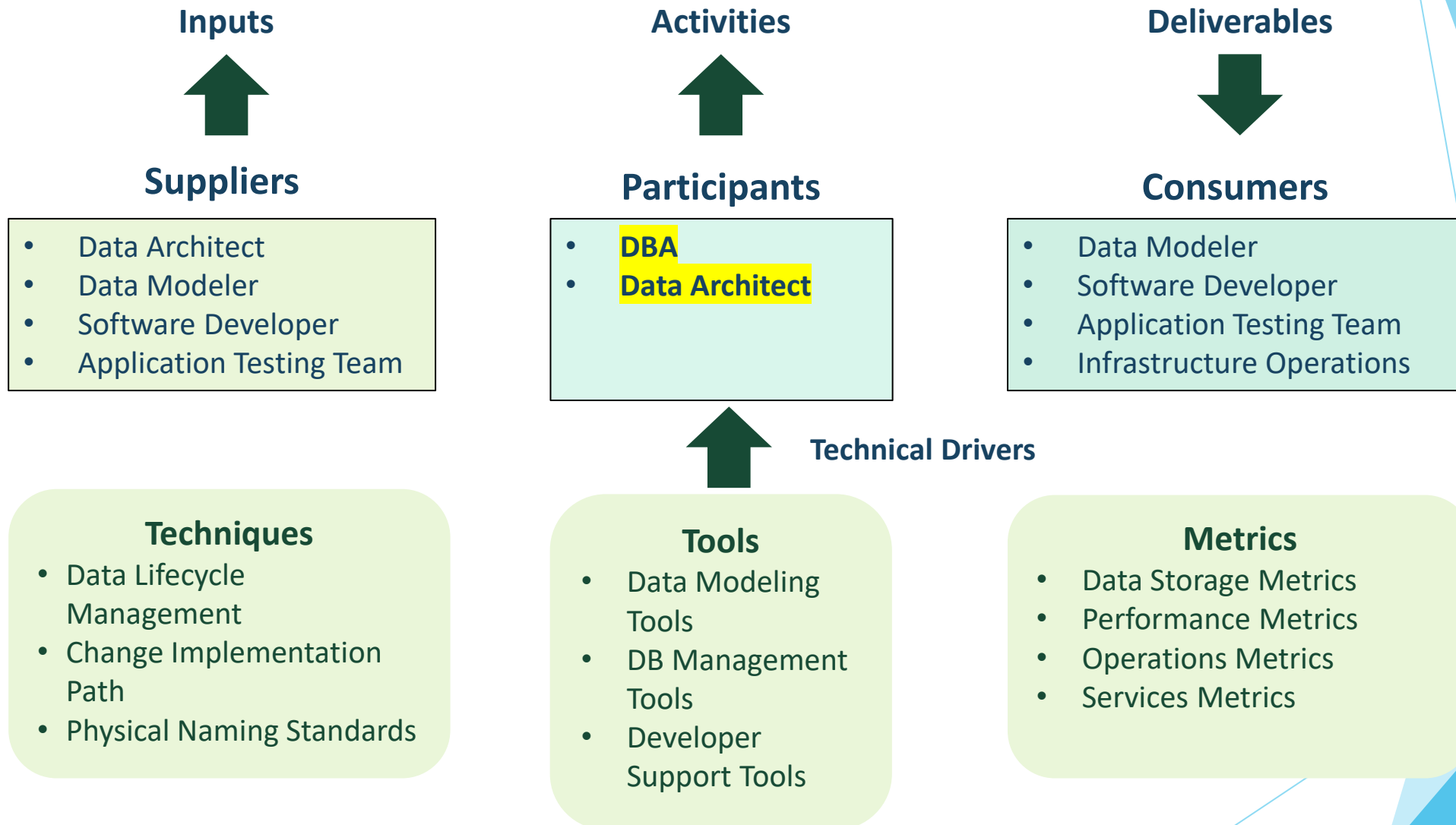
- Data Architecture
- Data Requirements
- Data Models
- SLA

Activities

1. Manage Database **Technology**
2. Manage Database **Operations**

Deliverables

1. **Business Continuity Plan**
2. Database **Environments**
3. Database **Performance**

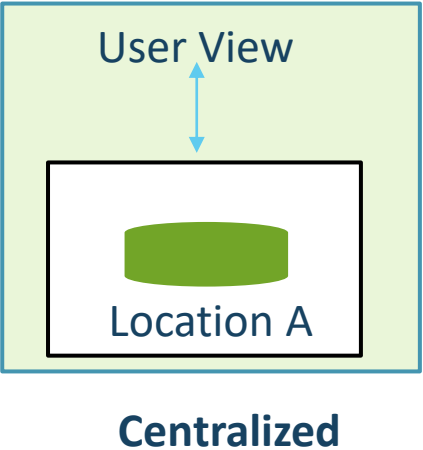


- Database Lifecycle Management
 - DBA is the **custodian** of all database changes and controlled environments
 - It includes implementing **policies and procedures** for
 1. data acquisition
 2. data migration
 3. data retention
 4. data expiration
 5. data disposition

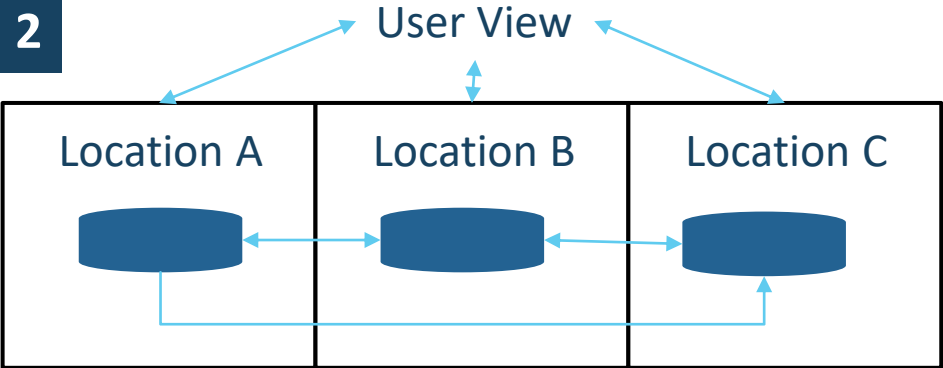
- **DBAs**
 - Critical role in both Data Storage & Operations and Data Security
 - **Supports development, test, QA, database environments**
 - DBAs, is a Data Team member. **Roles:**
 - Production DBA
 - Application DBA
 - Procedural and Development DBAs
 - NSA - Network Storage Administrators

Database Architecture Types

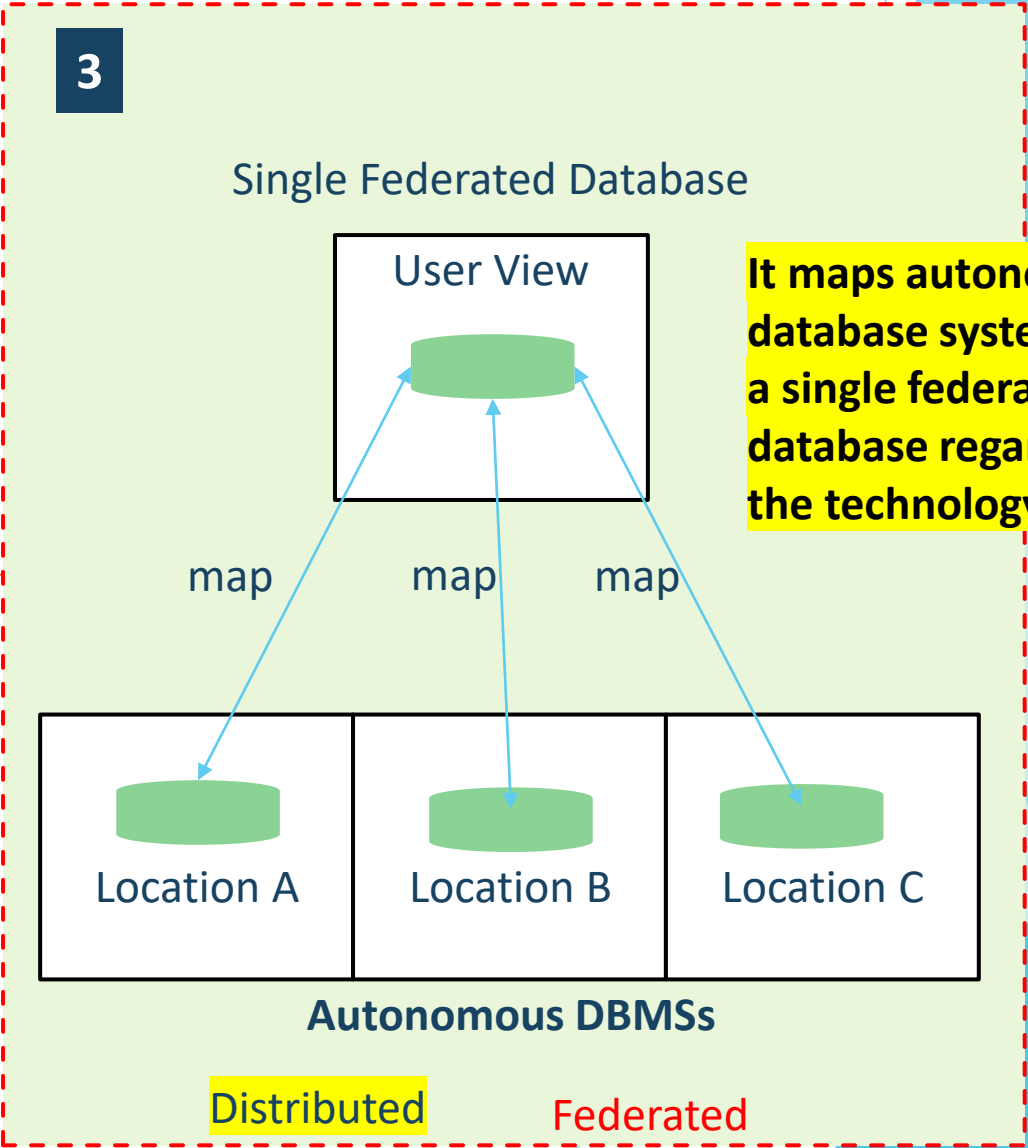
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- **Database Processing Types**

- 1. **ACID** **RDBMS**

- Atomicity
 - Consistency (w/all rules)
 - Isolation (independent trans.)
 - Durability

- 2. **BASE**

- Basically Available
 - Soft State
 - Eventual Consistency

- **Database Environments**
 - Pre-production Environments
 1. Development
 2. Testing (QA, User Acceptance, Performance)
 3. Sandboxes or Experimental
 - Production

- **Database Organization**

1. Hierarchical (tree schema)

2. **Relational**

- **Transactional (OLTP)**

- **Multidimensional (OLAP)**

- Temporal (valid time, transaction time, decision time)

3. Non-relational or NoSQL databases (Big Data)

- **GOVERNANCE**
 - Metrics
 - Data Storage metrics may include
 - Count of database types
 - Aggregated transaction statistics
 - Capacity metrics
 - Operational metrics may include
 - Service metrics
 - Information Asset Tracking to ensure compliancy
 - Data Audits and Data Validation vs acceptance criteria

